

1 Week 10

Definition 1.1. If $E \in \mathcal{E}$, $f : E \rightarrow \overline{\mathbb{R}}$ then we say that f is Lebesgue measurable if $f^{-1}(B)$ is Lebesgue measurable for all $B \in \mathcal{B}(\overline{\mathbb{R}})$.

Lemma 1.2. If $f : E \rightarrow \overline{\mathbb{R}}$ and if we define

$$\tilde{f}(x) = \begin{cases} f(x) & x \in E \\ 0 & x \notin E \end{cases}$$

then f is Lebesgue measurable if and only if $\tilde{f}$ is Lebesgue measurable.

Proof. If $B \in \mathcal{B}(\overline{\mathbb{R}})$ then if $0 \notin B$, $f^{-1}(B) = \tilde{f}^{-1}(B)$ and if $0 \in B$ then $\tilde{f}^{-1}(B) = f^{-1}(B) \cup E^c$, hence $f^{-1}(B) = \tilde{f}^{-1}(B) \cap E$. $\square$

Definition 1.3. We define $\mathbf{1}_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$. If it is Lebesgue measurable then so is E .

If $E \in \mathcal{E}$ we say that $\varphi : E \rightarrow \overline{\mathbb{R}}$ is simple if $\varphi = \sum_{k=1}^n \alpha_k \mathbf{1}_{E_k}$ where $E_k \subseteq E$ are Lebesgue measurable and $\alpha_k \in \mathbb{R}$.

Note that you can always rewrite the representative so that the E_k are disjoint and the α_k 's are unique. This representation is called canonical.

Lemma 1.4. Let $f : E \rightarrow \mathbb{R}$ be a bounded Lebesgue measurable function with $E \in \mathcal{E}$, and let $\varepsilon > 0$. Then there exist simple functions φ_ε and ψ_ε such that $\varphi_\varepsilon \leq f \leq \psi_\varepsilon$ and $|\psi_\varepsilon(x) - \varphi_\varepsilon(x)| < \varepsilon$ for all $x \in E$.

Proof. Suppose that $M > |f(x)|$ for all $x \in E$. Let $y_i \in \mathbb{R}$ be an increasing sequence satisfying $y_1 = -M$ and $y_n = M$. Then we let $E_k = f^{-1}((y_k, y_{k+1}))$ for all $1 \leq k \leq n-1$. We set $\varphi = \sum_{k=1}^{n-1} y_k \mathbf{1}_{E_k}$ and $\psi = \sum_{k=1}^{n-1} y_{k+1} \mathbf{1}_{E_k}$. By definition $\varphi \leq f \leq \psi$ and $|\varphi - \psi| \leq \max_k |y_{k+1} - y_k|$. By choosing y_k to be close enough, we get the second desired condition. $\square$

Theorem 1.5. Let $f : E \rightarrow \overline{\mathbb{R}}$ be Lebesgue measurable. Then

- (i) If $f \geq 0$ then there exists an increasing sequence of simple functions $\varphi_n \geq 0$ such that $\lim_{n \rightarrow \infty} \varphi_n(x) = f(x)$ for all $x \in E$.
- (ii) In general, there exists a sequence of simple functions φ_n such that $\lim_{n \rightarrow \infty} \varphi_n(x) = f(x)$ and $|\varphi_n(x)| \leq |f(x)|$ for all $x \in E$.

Proof. (ii) follows from (i) by writing $f = f_+ - f_-$ and finding simple ψ_n, χ_n that approximate f_+, f_- in the way of (i) and defining $\varphi_n = \psi_n - \chi_n$. Since at most one of f_+, f_- is non-zero at a given x , the same is the case for ψ_n, χ_n and hence the absolute value condition holds. The pointwise convergence obviously holds as well.

For (i), if $f \geq 0$, if we set $f_n = \max\{f, n\}$ then by the previous lemma, there is an increasing sequence $\psi_n \leq f_n$ and $|f_n - \psi_n| < 1/n$ for all n . Then we take $\varphi_1 = \max\{\psi_1, 0\}$ and $\varphi_k = \max\{0, \psi_k, \varphi_{k-1}\}$ for $k \geq 2$. By induction, we find that φ_k 's are step functions such that $0 \leq \varphi_k \leq \varphi_{k+1} \leq f_k$, and $|f_k - \varphi_k| < 1/k$. With this we get $\varphi_n(x) \rightarrow f(x)$ for all $x \in E$. $\square$

Theorem 1.6 (Egorov). Let $f_n : E \rightarrow \mathbb{R}$ be a sequence of Lebesgue measurable functions converging pointwise to $f : E \rightarrow \mathbb{R}$. We also assume that $m(E) < \infty$. Then for all $\varepsilon > 0$, there is a closed set $F \subseteq E$ such that $m(E \setminus F) < \varepsilon$ and $f_n \rightarrow f$ converges uniformly on F (that is, for all $\delta > 0$ there is some N such that $|f(x) - f_n(x)| < \delta$ for all $x \in F$ and $n \geq N$).

Lemma 1.7. In the setting of the theorem, for all $u > 0$ and $\delta > 0$ there is some $A \subseteq E$ and an $N \geq 0$ such that $m(E \setminus A) < \delta$ and for all $x \in A$ and $n \geq N$ we have $|f_n(x) - f(x)| < u$.

Proof. Set $E_n = \{x \in E : |f_n(x) - f(x)| < u, \forall k \geq n\} = \bigcap_{k=n}^{\infty} \{x : |f_k(x) - f(x)| < u\}$. Since $f_n \rightarrow f$ pointwise, $E = \bigcup_{n=1}^{\infty} E_n$. Furthermore, E_n are increasing ($E_n \subseteq E_{n+1}$ for all n). We then have $m(E) = \lim_{n \rightarrow \infty} m(E_n)$. Since $m(E) < \infty$, there is some N such that $|m(E) - m(E_N)| < \delta$, and hence $m(E \setminus E_N) = m(E) - m(E_N) < \delta$. Taking $A = E_N$ completes the proof. $\square$

Proof (Egorov). By the lemma, for all n there is some A_n such that $m(E \setminus A_n) < \varepsilon/2^n$ and there exists $M(n)$ such that $|f(x) - f_k(x)| < 1/n$ for all $k \geq M(n)$. Setting $A = \bigcup_{n=1}^{\infty} A_n$, we have

$$\begin{aligned} m(E \setminus A) &= m(E \cap A^c) \\ &= m\left(E \cap \left(\bigcup_{n=1}^{\infty} A_n^c\right)\right) \\ &= m\left(\bigcup_{n=1}^{\infty} E \setminus A_n\right) \\ &\leq \sum_{n=1}^{\infty} m(E \setminus A_n) \\ &\leq \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} \\ &= \varepsilon \end{aligned}$$

Now if $x \in A$ then for all n , $x \in A_n$, so for all $k \geq M(n)$, we have $|f_k(x) - f(x)| < 1/n$. We thus see that $f_k \rightarrow f$ uniformly on A .

We now take F closed such that $m(A \setminus F) \leq \varepsilon$ with $F \subseteq A$ to get $m(E \setminus F) \leq m(E \setminus A) + m(A \setminus F) \leq 2\varepsilon$. $\square$

Theorem 1.8 (Lusin). Let $f : E \rightarrow \mathbb{R}$ be Lebesgue measurable with $m(E) < \infty$. Then for all $\varepsilon > 0$ there is a closed set F and a continuous $g : F \rightarrow \mathbb{R}$ such that $m(E \setminus F) < \varepsilon$ and $f = g$ on F .

Proposition 1.9. Let $\varphi : E \rightarrow \mathbb{R}$ be a simple function and let $\varepsilon > 0$. Then there exists a closed $F \subseteq E$ and a continuous $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $m(E \setminus F) < \varepsilon$ and $g = \varphi$ on F .

Proof. Write $\varphi = \sum_{k=1}^{\infty} \alpha_k \mathbb{1}_{E_k}$ with E_k being disjoint. Take F_k closed and bounded such that $F_k \subseteq E_k$ with $m(E_k \setminus F_k) < \varepsilon/n$. Since F_k 's are disjoint compact sets, we have $\delta = \min_{i \neq j} \inf_{x \in F_i, y \in F_j} |x - y| > 0$. For any set A , define $d(x, A) = \inf_{y \in A} |x - y|$.

Exercise 1.10. Show that $x \mapsto d(x, A)$ is continuous.

Proof. TODO $\square$

Let $\rho(x) : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ as 1 on $[0, \delta/3]$, 0 on $[2\delta/3, \delta]$, and linear on $[\delta/3, 2\delta/3]$ such that ρ becomes connected. Define $g_i = \alpha_i \rho(d(x, F_i))$. Each g_i is continuous and if $x \in F_i$ then $\rho(d(x, F_i)) = 1$ and if $x \in F_j$ for $j \neq i$ then $\rho(d(x, F_j)) = 0$ because $d(x, F_i) \geq \delta$. We then define $g(x) = \sum_{i=1}^n g_i(x)$.

Define $F = \bigcup_{n=1}^{\infty} F_n$, and note that $g(x) = \varphi(x)$ for $x \in F$ and $m(E \setminus F) \leq \sum_{k=1}^n m(E_k \setminus F_k) \leq n\varepsilon/n = \varepsilon$.

Lemma 1.11. If E has finite measure then for any $\varepsilon > 0$ there is some compact K such that $m(E \setminus K) < \varepsilon$.

Proof. Set $E_n = E \cap [-n, n]$. Then since $\bigcup_n E_n = E$ and $m(E) = \lim_{n \rightarrow \infty} m(E_n)$, we can find E_k such that $m(E \setminus E_k) < \varepsilon$. $\square$

Then take $F \subseteq E_k$ with F closed and $m(E_k \setminus F) < \varepsilon \Rightarrow m(E \setminus F) \leq m(E \setminus E_k) + m(E_k \setminus F) \leq 2\varepsilon$. $\square$

Proof (Lusin). Take φ_n simple functions such that φ converge to f pointwise and let g_n be a sequence of continuous functions such that $g_n = \varphi_n$ on a closed set F_k with $m(E \setminus F_k) \leq \varepsilon/2^n$. By Egorov, there is a closed $G \subseteq E$ with $m(E \setminus G) < \varepsilon$ such that $\varphi_n \rightarrow f$ uniformly on G . Take $F = (\bigcap_{n=1}^{\infty} F_n) \cap G$. Then F is closed and $m(E \setminus F) \subseteq m(E \setminus G) + \sum_n m(E \setminus F_n) \leq 2\varepsilon$. On F , we have $g_n = \varphi_n$ for all n , so $g_n \rightarrow f$ uniformly on F . Thus the function $g(x) = \lim_{n \rightarrow \infty} g_n(x)$ for $x \in F$ is continuous. $\square$

2 Week 11

Definition 2.1. We define

$$\int_E \sum_{j=1}^n \alpha_j \mathbb{1}_{E_j} = \sum_{j=1}^n \alpha_j m(E_j)$$

and

$$\int_E f = \sup \left\{ \int_E \varphi : \varphi \leq f, \varphi \text{ simple} \right\}$$

Theorem 2.2 (Monotone Convergence Theorem). Let $f_n : E \rightarrow [0, \infty]$ be measurable, $f_n \leq f_{n+1}$ for all n . Let $f = \lim_{n \rightarrow \infty} f_n$ pointwise. Then $\lim_{n \rightarrow \infty} \int_E f_n = \int_E f$.

Proof. Since $f_n \leq f$, we have $\int_E f_n \leq \int_E f \Rightarrow \lim_{n \rightarrow \infty} \int_E f_n \leq \int_E f$. Let $0 < \alpha < 1$ and let $\varphi : E \rightarrow [0, \infty)$ be simple with $\varphi \leq f$. Suffices to show that $\lim_{n \rightarrow \infty} \int_E f_n \geq \alpha \int_E \varphi$. Let $E_k = \{x \in E : f_k(x) \geq \alpha \varphi(x)\}$ and note that $E_k \subseteq E_{k+1}$ and $E = \bigcup_{k=1}^{\infty} E_k$. We then have

$$\lim_{n \rightarrow \infty} \int_E f_n \geq \int_E f_k \geq \int_E \mathbb{1}_{E_k} f_k \geq \int_E \mathbb{1}_{E_k} \alpha \varphi = \alpha \int_E \mathbb{1}_{E_k} \varphi$$

Writing $\varphi = \sum_{j=1}^n \alpha_j \mathbb{1}_{F_j}$, we have

$$\alpha \int_E \mathbb{1}_{E_k} \varphi = \alpha \sum_{j=1}^n \alpha_j m(F_j \cap E_k) \leq \lim_{n \rightarrow \infty} \int_E f_n$$

This holds for all k so take $k \rightarrow \infty$. Since $F_j \cap E_k \subseteq F_j \cap E_{k+1}$ for all k and $\bigcup_{k=1}^{\infty} F_j \cap E_k = F_j$ so by continuity of measure $\lim_{n \rightarrow \infty} m(F_j \cap E_n) = m(F_j)$. Thus we have $\lim_{n \rightarrow \infty} \int_E f_n \geq \alpha \sum_{j=1}^n \alpha_j m(F_j) = \alpha \int_E \varphi$. Taking $\alpha \rightarrow 1$ and using the definition of Lebesgue integral, we see that $\lim_{n \rightarrow \infty} \int_E f_n \geq \int_E f$. Hence the equality holds. $\square$

Lemma 2.3. If $f, g : E \rightarrow [0, \infty]$ are measurable, then $\int_E f + g = \int_E f + \int_E g$.

Proof. Since f, g are nonnegative and measurable, there exist nonnegative simple functions φ_n, ψ_n that are increasing to f, g and $\varphi_n \rightarrow f, \psi_n \rightarrow g$ pointwise. Then

$$\int_E f + g = \int_E \lim_{n \rightarrow \infty} (\varphi_n + \psi_n) = \lim_{n \rightarrow \infty} \int_E \varphi_n + \psi_n = \lim_{n \rightarrow \infty} \int_E \varphi_n + \lim_{n \rightarrow \infty} \int_E \psi_n = \int_E f + \int_E g$$

$\square$

Corollary 2.4. If $f_n : E \rightarrow [0, \infty]$ are measurable then $\int_E \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} \int_E f_n$

Proof. Just apply MCT. $\square$

Proposition 2.5. Let $f : E \rightarrow [0, \infty]$ be measurable. Then $\int_E f = 0$ iff $f = 0$ a.e. on E .

Proof. If $f = 0$ a.e. on E then if $\varphi = \sum_{j=1}^n \alpha_j \mathbb{1}_{E_j}$ with E_j disjoint, $\alpha_j \geq 0$, satisfies $\varphi \leq f$, then if $\alpha_j > 0$, we have $m(E_j) = 0$. Thus $0 \leq \int_E f \leq \int_E \varphi \leq 0$ so $\int_E f = 0$.

If $\int_E f = 0$, consider the set $f^{-1}((0, \infty]) = \bigcup_{k=1}^{\infty} f^{-1}((1/k, \infty])$, and define $E_k := f^{-1}((1/k, \infty])$. We then have

$$0 = \int_E f \geq \int_E \mathbb{1}_{E_k} f \geq \int_E \frac{1}{k} \mathbb{1}_{E_k} = \frac{m(E_k)}{k} \geq 0$$

Thus $m(E_k) = 0$ so $m(f^{-1}((0, \infty])) = \lim_{n \rightarrow \infty} m(E_k) = 0$. Since the set on which $f > 0$ is of measure zero, we have $f = 0$ a.e. on E . $\square$

Theorem 2.6 (Better MCT). Let $f_n, f : E \rightarrow [0, \infty]$ be measurable. Assume that $f_n \leq f_{n+1}$ a.e. and assume $f = \lim_{n \rightarrow \infty} f_n$ a.e. on E . Then $\lim_{n \rightarrow \infty} \int_E f_n = \int_E f$

Proof. Let $E_0 = \{x \in E : \lim_{n \rightarrow \infty} f_n(x) = f(x), f_n(x) \leq f_{n+1}(x)\}$. Then $m(E \setminus E_0) = 0$ ($E \setminus E_0$ is a union of sets of measure zero). Now, we have

$$\lim_{n \rightarrow \infty} \int_E f_n = \lim_{n \rightarrow \infty} \int_E f_n (\mathbb{1}_{E_0} + \mathbb{1}_{E \setminus E_0}) = \lim_{n \rightarrow \infty} \int_E f_n \mathbb{1}_{E_0} + \int_E f_n \mathbb{1}_{E \setminus E_0} = \int_E \lim_{n \rightarrow \infty} f_n \mathbb{1}_{E_0} = \int_{E_0} f = \int_E f$$

$\square$

Theorem 2.7 (Fatou's lemma). If $f_n : E \rightarrow [0, \infty]$ are measurable then

$$\int_E \liminf_{n \rightarrow \infty} f_n \leq \liminf_{n \rightarrow \infty} \int_E f_n.$$

In particular, if $f_n \rightarrow f$ a.e. then $\int_E f \leq \liminf_{n \rightarrow \infty} \int_E f_n$.

Proof.

$$\int_E \liminf_{n \rightarrow \infty} f_n = \int_E \lim_{n \rightarrow \infty} \left(\inf_{k \geq n} f_k \right) = \lim_{n \rightarrow \infty} \int_E \inf_{k \geq n} f_k \leq \lim_{n \rightarrow \infty} \inf_{k \geq n} \int_E f_k = \liminf_{n \rightarrow \infty} \int_E f_n$$

$\square$

Theorem 2.8 (Markov's inequality). If $f : E \rightarrow [0, \infty]$ is measurable then for all $\alpha > 0$, $m(f^{-1}([\alpha, \infty])) \leq \frac{1}{\alpha} \int_E f$.

Proof. Let $F = f^{-1}([\alpha, \infty])$, then $f(x)/\alpha \geq 1$ on F , so

$$m(F) = \int_E \mathbb{1}_F \leq \int_E \mathbb{1}_F \frac{f}{\alpha} \leq \frac{1}{\alpha} \int_E f$$

$\square$

Corollary 2.9. If $t > 0$, $m(f^{-1}([\alpha, \infty])) \leq e^{-t\alpha} \int_E e^{tf}$.

Proof. Holds since $m(f^{-1}([\alpha, \infty])) = m((e^{tf})^{-1}([\alpha, \infty]))$. $\square$

Corollary 2.10. If $\int_E f < \infty$ then $f(x)$ is finite a.e.

Proof. $m(f^{-1}(\infty)) \leq m(f^{-1}([n, \infty])) \leq \frac{1}{n} \int_E f \rightarrow 0$. $\square$

2.1 Integral for general $f : E \rightarrow \overline{\mathbb{R}}$

Definition 2.11. We say that a measurable $f : E \rightarrow \overline{\mathbb{R}}$ is integrable if $\int_E f_+ + \int_E f_- = \int_E |f| < \infty$. If f is integrable then we define $\int_E f = \int_E f_+ - \int_E f_-$.

Lemma 2.12. Integrable functions form a vector space in that if f, g are integrable functions then so is $f + g$ and if f is integrable then so is λf for any $\lambda \in \mathbb{R}$.

Proof. Note that $\int_E |f + g| \leq \int_E |f| + \int_E |g| < \infty$ and $\int_E |\lambda f| = |\lambda| \int_E |f| < \infty$ which proves the lemma. $\square$

Exercise 2.13. Check that $\int_E \alpha f + \beta g = \alpha \int_E f + \beta \int_E g$ for all integrable f, g and constants α, β .

Exercise 2.14. Check that Fatou's lemma holds if $f_n \geq 0$ a.e.

Theorem 2.15 (Dominated Convergence Theorem). Let $f_n : E \rightarrow \overline{\mathbb{R}}$ be a sequence of measurable functions that converge pointwise a.e. to some $f : E \rightarrow \overline{\mathbb{R}}$. Assume that there is an integrable g such that $|f_n| \leq g$ a.e. Then $\lim_{n \rightarrow \infty} \int_E f_n = \int_E f$.

Proof. By assumptions, $g + f_n, g - f_n$ are nonnegative a.e. so we can apply Fatou's lemma. This gives us

$$\int_E g + \int_E f = \int_E g + f = \int_E \lim_{n \rightarrow \infty} (g + f_n) \leq \liminf_{n \rightarrow \infty} \int_E g + f_n = \int_E g + \liminf_{n \rightarrow \infty} \int_E f_n$$

Similarly, we have

$$\int_E g - \int_E f \leq \int_E g + \liminf_{n \rightarrow \infty} \int_E -f_n = \int_E g - \limsup_{n \rightarrow \infty} \int_E f_n$$

Since $\int_E g = \int_E |g| < \infty$, we get

$$\int_E f \leq \liminf_{n \rightarrow \infty} \int_E f_n \leq \limsup_{n \rightarrow \infty} \int_E f_n \leq \int_E f$$

Thus we have $\lim_{n \rightarrow \infty} \int_E f_n = \int_E f$. $\square$

Example 2.16. If E satisfies $m(E) < \infty$ and there is $B > 0$ s.t. $|f_n| \leq B$ a.e. then $\int_E \lim_{n \rightarrow \infty} f_n = \lim_{n \rightarrow \infty} \int_E f_n$ by DCT with $g = B$.

Definition 2.17. If $f, g : E \rightarrow \overline{\mathbb{R}}$ are measurable we say that $f \sim g$ if $f = g$ a.e. This is an equivalence relation. Let $\mathcal{F}$ be the set of equivalence classes and define $\alpha[f] + \beta[g] = [\alpha f + \beta g]$. It is easy to check that this gives a well-defined definition of addition and scalar multiplication. For $1 \leq p < \infty$, we define $L^p(E)$ to be the set of all $[f] \in \mathcal{F}$ such that $\|f\|_{L^p(E)} := \left(\int_E |f|^p\right)^{1/p} < \infty$.